

A Rare Occurrence of Rotational Retro-Esophageal Gastric Body Herniation Through a Nissen Fundoplication.

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Anti-reflux procedures have become a mainstay in managing gastroesophageal reflux disease (GERD) and hiatal hernia. Unfortunately, post-operative events such as breakdown of the wrap, downward slippage, or transdiaphragmatic herniation of an intact wrap cause these procedures to fail and create complications such as recurrent hiatal hernia and reflux dysphagia, regurgitation, and obstruction requiring revision surgery. We discuss a case of a rotational retro-esophageal herniation of the gastric body through a Nissen fundoplication presenting as obstruction, dysphagia, and regurgitation, highlighting the peculiar nature of this presentation and the ease of misdiagnosis given its rarity.

Nissen fundoplication has a vagolytic effect on the lower esophageal sphincter.

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Laparoscopic Nissen fundoplication is an increasingly utilized option for the treatment of gastroesophageal reflux disease (GERD). However, many questions remain as to the mechanism by which this operation prevents GERD in those without hiatal hernias or incompetent lower esophageal sphincters (LES). It is known that these patients experience reflux due to excess transient lower esophageal sphincter relaxations (TLESR), inappropriate and short-lived relaxation of the LES and crural diaphragm. The purpose of this study was to determine if Nissen fundoplication affects the neural pathways involved in the TLESR reflex. Five dogs were anesthetized and intubated. Both vagal nerves and the right phrenic nerve were isolated. A continuous water perfusion manometric catheter was situated at the LES. The nerves were then electrically stimulated and the resultant pressure at the LES measured at baseline, and during and after an open Nissen fundoplication. The mean LES pressures before dissection, after esophago-gastric mobilization, and after fundoplication were 47 +/- 13 mmHg, 21 +/- 9 mmHg, and 14 +/- 4 mmHg, respectively. All differences were significant. There was no change noted in LES pressure with stimulation of either or both of the phrenic nerves without concomitant vagal stimulation. Nissen fundoplication may prevent GERD in those without a hiatal hernia or incompetent LES by disrupting the efferent vagal fibers to the LES. Such fibers mediate TLESR which are responsible for GERD in these patients.

LATE EVALUATION OF PATIENTS OPERATED FOR GASTROESOPHAGEAL REFLUX DISEASE BY NISSEN FUNDOPLICATION.

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Surgical treatment of GERD by Nissen fundoplication is effective and safe, providing good results in the control of the disease. However, some authors have questioned the efficacy of this procedure and few studies on the long-term outcomes are available in the literature, especially in Brazil. To evaluate patients operated for gastro-esophageal reflux disease, for at least 10 years, by Nissen fundoplication. Thirty-two patients were interviewed and underwent upper digestive endoscopy, esophageal manometry, 24 h pH monitoring and barium esophagogram, before and after Nissen fundoplication. Most patients were asymptomatic, satisfied with the result of surgery (87.5%) 10 years after operation, due to better

symptom control compared with preoperative and, would do it again (84.38%). However, 62.5% were in use of some type of anti-reflux drugs. The manometry revealed lower esophageal sphincter with a mean pressure of 11.7 cm H₂O and an average length of 2.85 cm. The average DeMeester index in pH monitoring was 11.47. The endoscopy revealed that most patients had a normal result (58.06%) or mild esophagitis (35.48%). Barium swallow revealed mild esophageal dilatation in 25,80% and hiatal hernia in 12.9% of cases. After at least a decade, most patients were satisfied with the operation, asymptomatic or had milder symptoms of GERD, being better and with easier control, compared to the preoperative period. Nevertheless, a considerable percentage still employed anti-reflux medications.

Vanishing spleen after Nissen fundoplication: a case report.

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Nissen fundoplication is a generally accepted treatment for severe gastro-oesophageal reflux after conservative management has failed. The surgical techniques and the complications that may develop following the operation have been well described. However, necrosis of the spleen is a rare complication. We report here a patient with Down syndrome with a vanishing spleen after a Nissen fundoplication, who died of overwhelming pneumococcal septic shock 7 months after the operation. Vascular anomaly in Down syndrome, inadvertent ligation of the splenic artery or volvulus of the spleen may have caused a compromised splenic arterial circulation. Conclusion Nissen fundoplication may be associated with vanishing spleen and, consequently, with devastating consequences.

Laparoscopic Toupet versus Nissen fundoplication for the treatment of gastroesophageal reflux disease.

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The Nissen fundoplication is the most popular laparoscopic operation performed for the surgical treatment of gastroesophageal reflux disease (GERD). However, for patients in whom esophageal peristalsis is documented to be weak preoperatively, use of a partial wrap, or Toupet procedure, has often been used as an alternative to lessen the potential for postoperative dysphagia. Recent reports have criticized the Toupet procedure as having a higher long-term failure rate than the Nissen approach, especially for patients with severe forms of GERD. We reviewed our experience performing laparoscopic antireflux surgery over a 7-year period and compared the results of patients undergoing laparoscopic Nissen versus Toupet procedures. All procedures were performed at our institution by a single surgeon. Data recorded included preoperative demographic data, preoperative disease parameters, perioperative data, postoperative course, and symptom scores. Follow-up was based on a combination of medical records and phone interviews. There were 142 patients with complete records allowing review for this study. Of these, 118 underwent 122 Nissen funduplications and 26 underwent 27 Toupet funduplications. Selection of the procedure was based on preoperative manometric studies. There were seven reoperations. Seven of the patients (28%) who underwent Toupet procedures had severe GERD, a percentage comparable to the Nissen group (31.6%). Preoperative parameters were comparable for both groups, although the Toupet patients had lower average preoperative LES pressures (9.79 mmHg) than did the Nissen patients (16.1 mmHg, $P < 0.05$). The operative duration, operative blood loss, morbidity, length of hospitalization, need for reoperation, and efficacy in terms of relieving symptoms (average

follow-up = 27.5 months) were comparable for both groups. Based on this experience, the Toupet procedure seems safe and effective in treating the symptoms of GERD, including patients with severe forms of the disease. We recommend its selective use in patients with preoperative esophageal hypomotility who are undergoing laparoscopic antireflux surgery.

Laparoscopic Nissen fundoplication at a teaching center: prospective analysis of 103 consecutive patients.

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The Nissen fundoplication is the most extensively studied and successfully employed surgical solution to gastroesophageal reflux disease (GERD). Early success with the application of minimally invasive techniques to this procedure has been reported by several authors. One hundred three consecutive patients were operated on for the symptoms and complications of GERD. Preoperative evaluation consisted of esophagogastroduodenoscopy and esophageal manometry. Twenty-four-hour esophageal pH was obtained selectively. All cases were performed in a traditional training environment, adhering to techniques previously described in the open literature. Clinical data consisted of operative time, postoperative hospital days, days to resumption of normal activities, and morbidity. Patients were followed clinically for the incidence of dysphagia, bloating, and recurrent reflux symptoms. These were graded using a modified Visick score prior to discharge, at 1, 3, and 6 months, and then annually. All patients underwent completion of their procedure; however, four required conversion to open technique and were excluded from analysis. Mean operative time for the 99 laparoscopic procedures was 180 min. Mean operative time was significantly longer for the first 50 cases (202.1 min) than the last 49 (164.2 min). Mean postoperative hospital stay was 2.3 days with 10 days to resumption of normal activities. Mean follow-up was 15 months (range 3-39 months). Three of the four treatment failures underwent open revision with good subsequent results. Patient satisfaction as reflected by the modified Visick score reveals 96% good to excellent results (Visick 1 or 2) with no persistent dysphagia. The Nissen fundoplication can be safely performed using minimally invasive techniques with the benefit of postoperative recovery typical of other laparoscopic procedures. By strictly adhering to the primary technical principles previously described in the open approach, early results are comparable. The procedure can be safely performed in a traditional training environment. The modified Visick system is a simple and effective method to quantify postoperative patient satisfaction.

The histological appearances of Nissen-type fundoplication in the ferret.

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Nissen fundoplication is of proven effectiveness in the surgical control of gastro-oesophageal reflux. However, our understanding of the effects of fundoplication upon foregut physiology is incomplete and post-operative symptoms are often poorly understood. This experimental study aimed systematically to characterize the tissue response to fundoplication in an animal model, to improve understanding of the effects of anti-reflux surgery upon foregut physiology. Nissen-type fundoplication was performed in the ferret, and the tissue response at 3 months examined histologically. Sham-operated animals that underwent laparotomy but no dissection or wrap, acted as controls. In fundoplicated animals, serosal fibrosis was observed in the gut wall, with patchy replacement of muscle by fibrous tissue. The ventral

and dorsal vagal nerve trunks were identified intact within the wrap. In cases where the wrap had spontaneously disrupted, fibrosis was more extensive and there was evidence of nerve damage. This is the first systematic description of the histopathological response to Nissen fundoplication. In the intact wrap, the vagal trunks appear spared, but there is fibrosis in the serosa, extending into the muscularis of the distal oesophagus and region of the cardia. These findings are discussed in relation to the effects of Nissen fundoplication upon gastric physiology and postoperative symptoms.

Dysphagia after Nissen fundoplication.

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Nissen fundoplication is a commonly used antireflux operation. After this operation symptoms such as dysphagia, inability to belch and vomit, and gas bloating are frequently reported in the literature. In 32 patients who underwent Nissen fundoplication 3.5-18 years ago, postprocedure dysphagia was studied using conventional manometry and 24 h ambulatory pressure and pH recording. Our study indicates that dysphagia tends to decrease after surgery. Neither conventional nor ambulatory pressure recording revealed motor abnormalities that could be held responsible for the dysphagia. Therefore, the data do not support the concept that dysphagia is a major complication of Nissen fundoplication.

Laparoscopic Nissen fundoplication: a prospective analysis of 200 consecutive patients.

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Nissen fundoplication is now the most common antireflux operation for gastroesophageal reflux disease. This study is a report on the laparoscopically performed floppy Nissen procedure. Two hundred consecutive patients were analyzed (84 women, 116 men, mean age 49 years, mean duration of symptoms 5 years) after laparoscopic Nissen fundoplication between 1992 and 1996. The main indications for surgery were daily heartburn, retrosternal pain, and regurgitation demanding continuous medical therapy. Eight patients (4%) had esophageal stricture, and 21 (11%) had Barrett's esophagus with intestinal metaplasia. All patients underwent upper gastrointestinal endoscopy, 24-h esophageal pH monitoring, and esophageal manometry before and 3 months after the operation. In addition, a questionnaire was completed an average of 2.2 years (range 1.0-4.6) after the operation. The results of the study were as follows: mortality was zero, and the morbidity rate was 5%. The mean hospital stay was 3.8 +/- 2.8 days, and sick leave was 14.3 +/- 10.4 days. Postoperatively, esophagitis was healed or significantly improved in all but 4 patients (98%), and 24-h pH and lower esophageal sphincter pressure were normal. After 2 years, 87% of the patients had Visick scores of I-II. It is concluded that laparoscopic floppy Nissen fundoplication provides an efficient and safe alternative for surgical treatment of gastroesophageal reflux disease.

Complications of the Nissen fundoplication.

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A more astute selection of patients and greater attention to technical details will preclude most

complications. The first step is standardizing the technique, with avoidance of unnecessary ancillary maneuvers. Equally important is a thorough preoperative evaluation. With these considerations in mind, the authors believe that Nissen fundoplication is the preferred method of treatment for patients with uncomplicated gastroesophageal reflux refractory to medical therapy.
